

BTEC Level 3 Applied Science • Unit 1 Biology • Answer sheet

Cell Structure — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

A–E

1. Nucleus: DNA / control of protein synthesis. Mitochondrion: aerobic respiration / ATP. Ribosome: protein synthesis.
2. Cell wall, chloroplasts, permanent vacuole.
3. Allows some substances to cross and restricts others.
- 4–5. Accept correct labels from the schematics: membrane, cytoplasm, nucleus, nucleolus, mitochondrion, RER/ribosomes, wall, vacuole, chloroplast. Wall freely permeable.
6. Phospholipid bilayer with proteins (and cholesterol / glycoproteins in animal cells); fluid mosaic.
7. Folds increase surface area for respiratory enzymes / electron-transport proteins.
8. High ATP demand for contraction.
9. Photosynthesis occurs in green shoots and leaves; roots absorb water and minerals.
10. Nucleus contains DNA and controls protein synthesis. Mitochondria carry out aerobic respiration / produce ATP. [4]
11. Many chloroplasts; position near the upper leaf; vacuole maintains turgidity; elongated shape. Any two explained. [4]
12. Wall: cellulose, freely permeable, support. Membrane: bilayer, selectively permeable. [3]
13. RER synthesises and processes proteins. Golgi modifies and packages them into vesicles for secretion.